⭐⭐⭐⭐⭐ 1. Make sense of, relate and use a variety of representations (Item 10)

Most central

- The problem provides:
  - A verbal description (diameter, height)
  - A diagram (river bed, highest point, centre M)
- Students must integrate:
  - The "highest point" shown in the diagram
  - The numerical information "150 m"
  - The geometric idea that the centre is one radius below the highest point

👉 This strongly reflects the ability to **integrate multiple representations**

⭐⭐⭐⭐⭐ 2. Develop mathematical diagrams ... and extract mathematical information from them (Item 6)

Very important

- Students do not need to draw a new diagram, but must:
  - Extract key geometric relationships from the given diagram
  - Recognise that the distance from the highest point to the centre equals the radius (**70 m**)
- The position of point M in the diagram conveys essential spatial information

👉 This demonstrates the ability to **extract mathematical information from a diagram**

⭐⭐⭐⭐ 3. Understand and utilise constructs based on definitions, rules and formal systems (Item 5)

Highly relevant

- Key geometric definitions include:
  - Diameter = 2 × radius
  - Distance from the centre of a circle to the highest point equals the radius
- Students must correctly apply these standard geometric rules

👉 This is a direct application of **basic geometric definitions**

⭐⭐⭐⭐ 4. Use an understanding of the context to guide the mathematical solving process (Item 13)

Important

- Students need to recognise that:
  - All heights are measured relative to the **river bed**
  - Information such as the boarding platform height and rotation direction is irrelevant to this question
- Contextual understanding helps students **filter relevant information**

👉 This reflects **context-guided mathematical reasoning**

⭐⭐⭐⭐ 5. Perform a simple calculation (Item 1)

Important but not dominant

- The calculations involved are:
  - Radius = 140 ÷ 2 = 70
  - Height of centre = 150 − 70 = **80**
- The arithmetic itself is straightforward

👉 The challenge lies in **setting up the correct relationship**, not in computation

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⭐⭐⭐ 6. Manipulate numbers and geometric representations (Item 7)

Moderately relevant

- The task involves:
  - Handling numerical data (140, 150)
  - Working with geometric representations (circle, radius, centre)

⭐⭐⭐ 7. Use a multi-step procedure leading to a solution (Item 12)

Moderately relevant

- At least two steps are required:
  1. Find the radius
  2. Subtract the radius from the highest point's height
- The procedure is linear and clear

⭐⭐ 8. Switch between different representations in the process of finding solutions (Item 11)

Lower relevance

- Students move from:
  - Diagram → geometric relationship → numerical calculation
- However, the representational shift is simple and direct

⭐⭐ 9. Implement a given strategy to determine a solution (Item 3)

Lower relevance

- No explicit strategy is provided in the problem
- Students rely on standard geometric reasoning

⭐⭐ 10. Select an appropriate strategy from a list (Item 2)

Lower relevance

- There is no comparison or selection among multiple strategies

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⭐ 11. Articulate a solution with intermediate results (Item 8)

Low relevance

- Only a final numerical answer is required
- No written explanation is demanded

⭐ 12. Make mathematical diagrams, graphs, or constructions (Item 4)

Low relevance

- A diagram is already provided
- Students are not required to construct their own

⭐ 13. Use mathematical tools, including technology (Item 9)

Minimal relevance

- No calculators or digital tools are needed

⭐ 14. Make generalisations based on results (Item 14)

Not relevant

- The problem focuses on a single specific situation
- No generalisation is expected

⭐⭐⭐⭐⭐ 1. Make sense of, relate and use a variety of representations（第10条）

最核心

- 题目同时提供：
  - 文字描述（直径、高度）
  - 示意图（河床、最高点、中心 M）
- 学生必须将：
  - 图像中的“最高点”
  - 数值信息“150 m”
  - 几何概念“中心在半径下方”
    进行整合理解

👉 这是**多重表征整合能力**的典型体现

⭐⭐⭐⭐⭐ 2. Develop mathematical diagrams ... and extract information from them（第6条）

非常关键

- 学生不需要画新图，但要：
  - 从现有图中识别关键几何关系
  - 明确“最高点到中心 = 半径 = 70 m”
- 图中 M 的位置提供了重要的空间信息

👉 体现了**从图中提取数学信息的能力**

⭐⭐⭐⭐ 3. Understand and utilise constructs based on definitions, rules and formal systems（第5条）

高度相关

- 关键数学定义：
  - 直径 = 2 × 半径
  - 圆心到最高点的距离等于半径
- 学生需要熟练运用这些基本几何规则

👉 属于**几何概念的直接应用**

⭐⭐⭐⭐ 4. Use an understanding of the context to guide the mathematical solving process（第13条）

重要

- 学生要意识到：
  - 所有高度都以“河床”为基准
  - 不需要考虑平台高度、转动方向等无关信息
- 情境理解帮助学生**筛选有用信息**

👉 这是**情境导向的数学理解**

⭐⭐⭐⭐ 5. Perform a simple calculation（第1条）

重要但不主导

- 计算步骤：
  - 半径 = 140 ÷ 2 = 70
  - 中心高度 = 150 − 70 = 80
- 运算本身非常简单

👉 难点不在计算，而在**建构正确关系**

⭐⭐⭐ 6. Manipulate numbers and geometric representations（第7条）

中等相关

- 涉及：
  - 数值处理（140、150）
  - 几何关系（圆、半径、中心）
- 无代数或复杂变换

⭐⭐⭐ 7. Use a multi-step procedure leading to a solution（第12条）

中等相关

- 至少包含两个步骤：
  1. 求半径
  2. 从最高点高度中减去半径
- 但步骤线性、清晰

⭐⭐ 8. Switch between different representations（第11条）

较低相关

- 从：
  - 图像 → 几何关系 → 数值计算
- 但表征切换较为自然，不复杂

⭐⭐ 9. Implement a given strategy（第3条）

较低相关

- 题目并未给出明确策略
- 学生基于经验自行完成

⭐⭐ 10. Select an appropriate strategy from a list（第2条）

较低相关

- 不涉及策略比较或选择

⭐ 11. Articulate a solution with intermediate results（第8条）

低相关

- 答案只需一个数值
- 不要求展示过程

⭐ 12. Make mathematical diagrams（第4条）

低相关

- 图已提供
- 不要求学生自行构图

⭐ 13. Use mathematical tools or technology（第9条）

几乎无关

- 不需要计算器、技术工具

⭐ 14. Make generalisations（第14条）

不相关

- 这是一个具体情境问题
- 不涉及规律推广